

Mount San Antonio College
Department of Natural Science

Predator/Prey Dynamics

Realistic Shark Behavior

ENGR 285 Differential Equations & Linear Algebra for Engineers

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Objective 1: The types of outcomes the simulation can have, and how common they appear to be. (Not just for the default parameters)

The simulation can have: Fish will breed and grow in time. Shark will consume fish and breed. More sharks means more fish are ate means less available fish for all sharks, sharks starve, sharks die. Less sharks means more fish will survive and breed, fish grow. Sharks grow because new fish are available. The cycle repeats until it becomes constant

Objective 2: The conditions (i.e. parameters) under which it appears the Lotka-Volterra Model models the simulation results well; i.e. when the populations appear to follow trajectories that resemble those of the Lotka-Volterra Model.

Objective 3: How you think the main three simulation parameters ("breed_time", "energy_gain", and "breed_energy") should relate to the four Lotka-Volterra parameters (α , β , δ , and γ), and why (both theoretically and in terms of the simulation results). *In order to compare the simulation data, recall that LVM solutions oscillate around particular population values for each species that are related to the LVM parameters.*

$$\frac{dx}{dt} = \alpha x - \beta xy$$

Where, α = *The natural per capita growth rate of the prey without predators.*

β = *The rate of predation or the effect of predators killing prey*

$$\frac{dy}{dt} = \delta xy - \gamma y$$

δ = *The efficiency of turning consumed prey into new predators.*

γ = *The natural per capita death rate (starvation rate) of the predators without prey.*

Objective 4: The difference in the simulation results for the two types of setups the program implements (when "basicSetup" is True or False). (Not just for the default parameters)

Objective 5 (special task): More realistic shark behavior: adjust the simulation so that sharks will not eat when their move counter is above a certain amount, OR so that sharks will avoid other sharks within a certain number of steps. Explore how the simulation results change under a variety of values for the move counter and/or avoidance preference size.

For our special task, we modified the code to add shark satiation as a new parameter to the model.

Conclusion